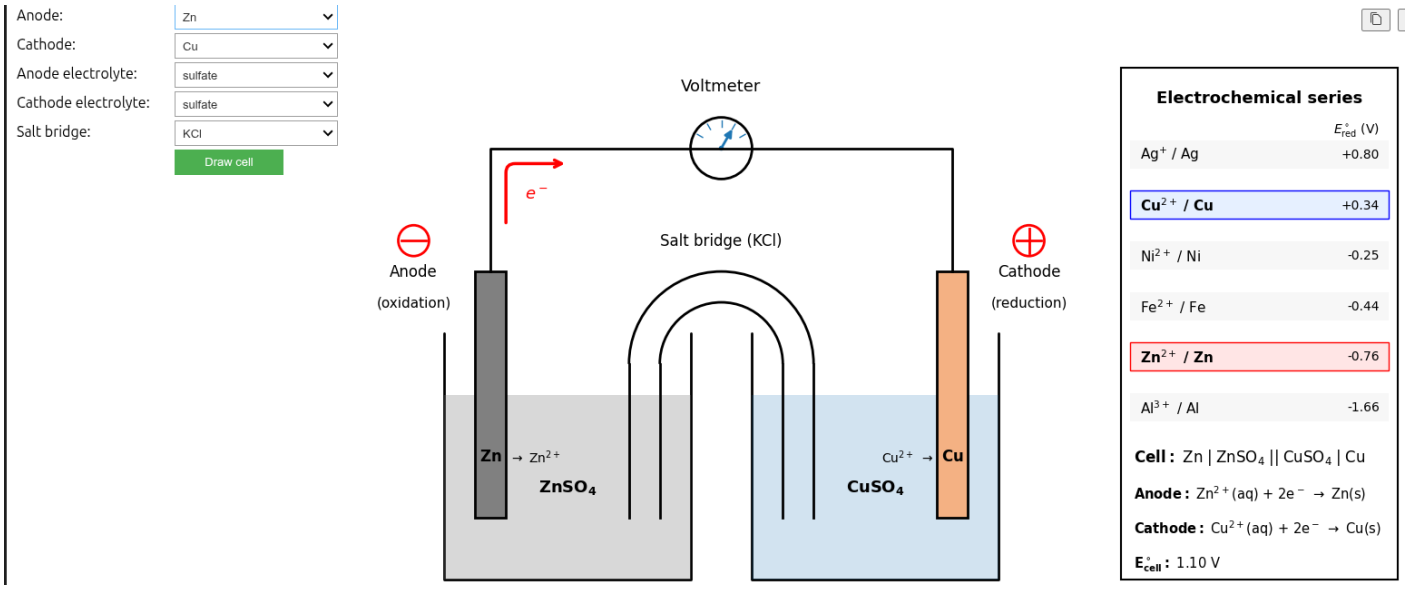

EXERCISE 1

Thermodynamics of electrochemical cells

April 15, 2026 | O. Schmidt | Theory and Computation of Energy Materials (IET-3)

JÜLICH Forschungszentrum

AIM OF TODAY'S EXERCISES

Today, we investigate the thermodynamic effects of electrochemical cells. The following link may be helpful

$\longrightarrow$ Exercises $\longleftarrow$

Objectives: The student is expected to be able to devise/ derive and/ or calculate

- The cell designation, the half cell reaction, the cell reaction, and the standard cell potential.

when provided with sufficient material.

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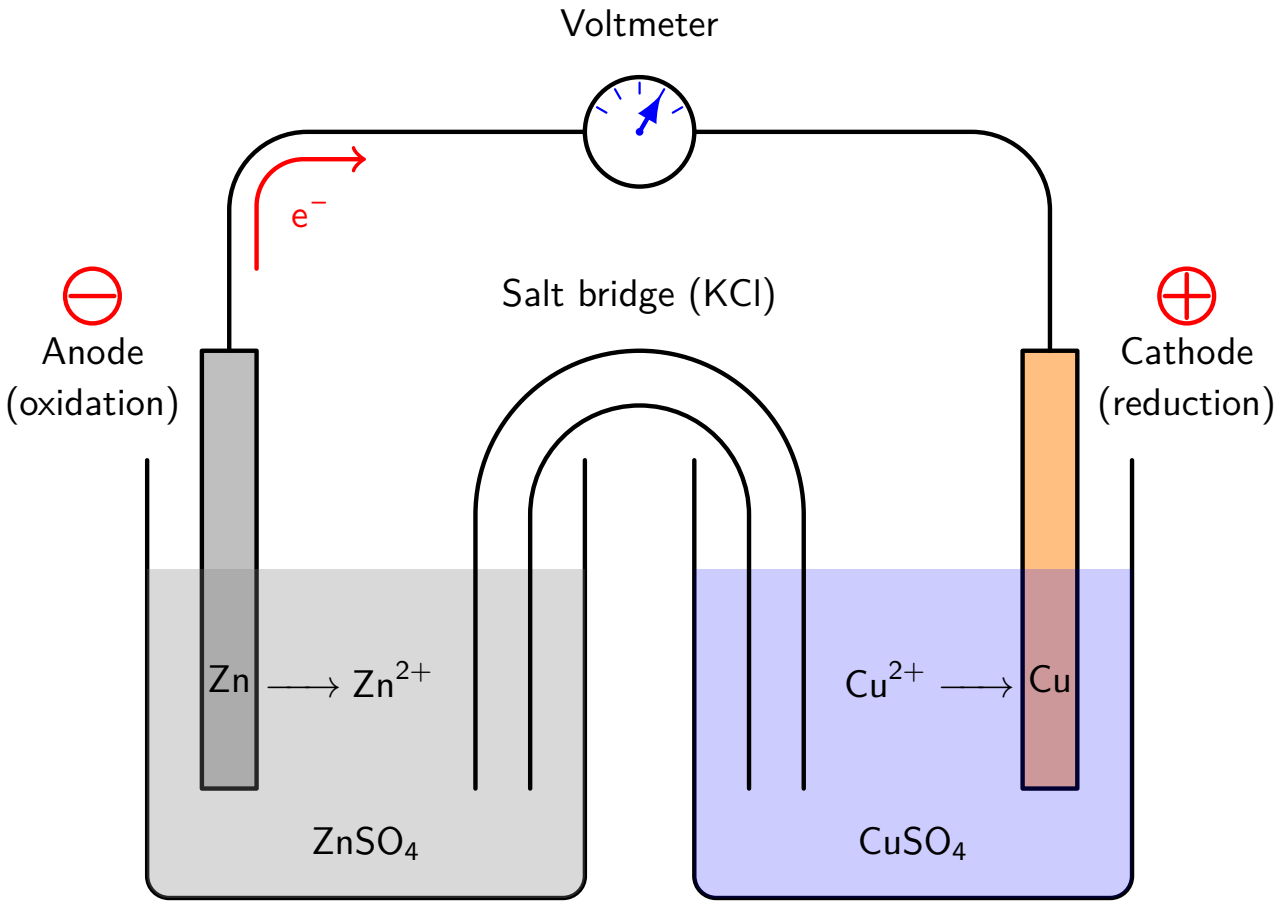
THE GALVANIC CELL: FIGURE



THE GALVANIC CELL: WORKED EXAMPLE

Cell designation: We have the following cell desingation

$$\mathrm{Zn(s)|ZnSO_4(aq)||CuSO_4(aq)|Cu(s)}$$

Half-cell reactions: The half-cell reactions are

$$\begin{aligned} \text{Anode}: \quad & \mathrm{Zn^{2+}\,(aq) + 2\,e^- \longrightarrow Zn\,(s)} \\ \text{Cathode}: \quad & \mathrm{Cu^{2+}\,(aq) + 2\,e^- \longrightarrow Cu\,(s)} \end{aligned}$$

Cell reaction: The cell reaction is found to be

$$\mathrm{Cu^{2+}\,(aq) + Zn\,(s) \longrightarrow Cu\,(s) + Zn^{2+}\,(aq)}$$

Standard cell potential: Using the Table on slide 5

$$E^{\ominus}_{\mathrm{cell}} = E^{\ominus}_{\mathrm{red}}\left(\mathrm{Zn^{2+}/Zn}\right) - E^{\ominus}_{\mathrm{red}}\left(\mathrm{Cu^{2+}/Cu}\right) = 0.34\,\mathrm{V} - (-0.76\,\mathrm{V}) = \underline{\underline{1.10\,\mathrm{V}}}$$

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THE GALVANIC CELL: QUESTIONS

1.1) For the following half cell reactions

$$2\,\mathrm{Ag}^{+}\,(\mathrm{aq}) + \mathrm{Ni}\,(\mathrm{s}) \longrightarrow 2\,\mathrm{Ag}\,(\mathrm{s}) + \mathrm{Ni}^{2+}\,(\mathrm{aq})$$

- 1.1a) Write the half-cell reactions.
- 1.1b) Calculate the standard cell potential.

1.2) Draw a diagram of the following galvanic cell with a salt bridge of your choice

$$\mathrm{Al}\,(\mathrm{s})\,|\mathrm{Al}\,(\mathrm{NO_3})_3\,(\mathrm{aq})\,||\mathrm{Fe}^{2+}\,(\mathrm{aq})\,|\mathrm{Fe}\,(\mathrm{s})$$

1.3) On slide 6, there is a diagram.

- 1.3a) Write the cell designation.
- 1.3b) Write the half-cell and cell reaction(-s).
- 1.3c) Calculate the standard cell potential.

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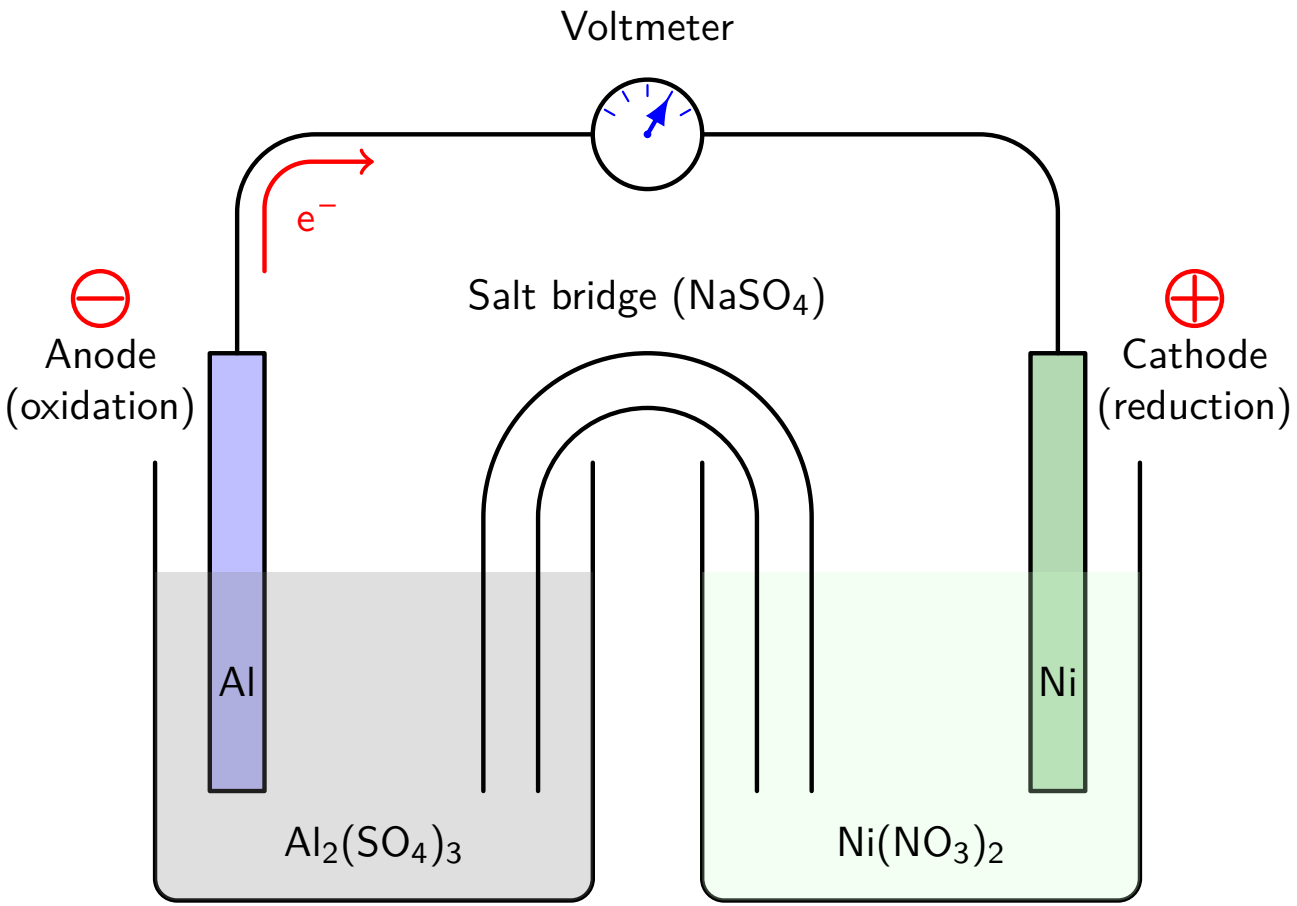
ELECTROCHEMICAL SERIES

Half-cell			$E^{\ominus}_{\mathrm{red}}$ **(V)**
$\mathrm{Ag^+(aq) + e^-}$	$\rightarrow$	$\mathrm{Ag(s)}$	$+0.80$
$\mathrm{Cu^{2+}(aq) + 2e^-}$	$\rightarrow$	$\mathrm{Cu(s)}$	$+0.34$
$\mathrm{Ni^{2+}(aq) + 2e^-}$	$\rightarrow$	$\mathrm{Ni(s)}$	-0.25
$\mathrm{Fe^{2+}(aq) + 2e^-}$	$\rightarrow$	$\mathrm{Fe(s)}$	-0.44
$\mathrm{Zn^{2+}(aq) + 2e^-}$	$\rightarrow$	$\mathrm{Zn(s)}$	-0.76
$\mathrm{Al^{3+}(aq) + 3e^-}$	$\rightarrow$	$\mathrm{Al(s)}$	-1.66

THE GALVANIC CELL: QUESTION 1.3